

FR250SDG Rear Helper Spring Kit Installation Manual

Model: FR250SDG (Rear Suspension Package)

Please read these instructions carefully before starting the installation. Ensure all standard safety precautions are observed while working underneath the vehicle.

1. Parts List

Before beginning, verify that all components listed below are present in your kit.

Ref #	Description / Specifications	Qty
1	Hex Cap Screw (HCS) (3/8" x 1" UNF, Grade 5, Zinc Plated)	2
2	Split Lock Washer (3/8", Grade 5, Zinc Plated)	4
3	Flat Washer (USS) (3/8" x 1", Grade 2, Zinc Plated)	2
4	Weldment Bracket (Painted)	2
5	Hollow Rubber Spring	2
6	Hex Cap Screw (HCS) (3/8" x 1-1/2" UNF, Grade 5, Zinc Plated)	2
7	Nylock Locknut (3/8" UNF, Grade 5, Zinc Plated)	2

2. Installation Instructions

1. Raise Vehicle: Elevate the vehicle using an appropriate lift or jack, ensuring the axle is allowed to hang freely. Secure the chassis adequately before proceeding with installation.

2. Remove OEM Components: Remove the vehicle's existing original equipment manufacturer (OEM) bump stop and its associated bushing.

3. Assemble Helper Spring Unit: Assemble the hollow rubber spring with the provided bracket.

4. Install Helper Spring Assembly: Mount the assembled helper spring unit to the vehicle frame, utilizing existing mounting holes and the fasteners supplied in this kit.

5. Lower Vehicle: Carefully lower the vehicle to the ground.

Installation Notes:

- Check for potential interference between the helper spring and any brake lines or cables.
- Utilize Nylock Locknut #7 with the bracket's offset holes as necessary to precisely center the helper spring over the axle.
- If the vehicle is equipped with an OEM peg-style casting, the hollow rubber spring is engineered to make contact with this component. Do not remove the OEM peg.

3. Clearance Instructions & Fitting Guidance

- **Recommended Clearance:** With the vehicle unloaded and at normal ride height, approximately **1/2" to 1-1/2" (13-38 mm)** of clearance between the rubber spring and its contact point is recommended.
- **Excessive Clearance:** A leveling kit, suspension lift, or other ride-height modification may increase the clearance. Excessive clearance may cause the helper spring to engage later, reducing its effectiveness in controlling sag and sway.
- **Insufficient Clearance:** Too little clearance or excessive preload may cause the helper spring to engage too early and may result in a firmer unloaded ride.
- **Under Load:** Cargo, accessories, or vehicle load may reduce or eliminate the clearance as the helper spring engages.
- **Note:** Actual clearance may vary depending on vehicle configuration, ride height, suspension condition, and load. If the unloaded clearance is significantly outside the recommended range, verify vehicle ride height, suspension modifications, load condition, and correct installation.

Important Safety Warnings:

- Do not exceed the vehicle manufacturer's recommended axle weight rating / GAWR.
- Inspect for potential interference between the hollow rubber spring and adjacent vehicle components. Under load, the rubber spring will expand in diameter.